Unsupervised Visual Anomaly Detection & Localization

AI-Powered Industrial Quality Inspection for High-Precision Manufacturing

AUTHORS

J. Ritte, L. Mushi, J. Juma, A. Mbwana, I. Sharif

DATASET

MVTec Anomaly Detection (3,260 Images)

Status: Supervised Baseline & Streamlit Live

THE MANUFACTURING BOTTLENECK

Manual Quality Checks vs. Production Scale

Human Inspection Limits

Slow visual fatigue and subjective evaluation lead to inconsistent surface defect detection.

Supervised ML Dilemma

Defective parts are extremely rare. Standard classifiers fail on unseen defect patterns.

High Cost of Missed Defects

Defects entering final assembly trigger severe component rejection and downtime costs.



METHODOLOGY ROADMAP

Two-Stage Anomaly Detection System

COMPLETED PHASE

Stage 3: Supervised ML Baseline

Engineered visual features (color, contrast, edge density) to train interpretable image-level classifiers.

Deliverable: Random Forest model & Streamlit web application.

NEXT MILESTONE

Stage 5: Unsupervised PatchCore

Deep feature embeddings via WideResNet-50 memory banks to locate exact pixel-level defects.

Deliverable: Zero-defect training & heatmaps.

DATA FOUNDATION

MVTec Anomaly Detection Benchmark

3,260
Cleaned Images

10
Object Categories

76%
Normal Parts (Label 0)

24%
Defective Parts (Label 1)

Objects & Textures

Covers industrial components like bottle, cable, capsule, metal nut, tile, grid, and screw.

Stratified Train/Test Split

80% training set (2,608 images) and 20% test set (652 images) preserving exact class ratios.

DATA PREPROCESSING

Extracting Visual Signal from Images

1

Resize & Standardize

Scale images to fixed 128×128 px resolution.

2

Color Statistics

Calculate RGB mean and standard deviation vectors.

3

Edge & Contrast

Compute Canny edge density and grayscale brightness.

4

Encode & Scale

One-hot encode categories to assemble 21 predictor features.

Data Hygiene Guarantee: Feature scaler fitted strictly on the training set to prevent test data leakage.

MODEL EVALUATION

Supervised Baseline Performance

Model Architecture	Accuracy	Precision (Anomaly)	Recall (Anomaly)	F1-Score (Anomaly)
Random Forest (Selected)	**0.853**	**0.729**	0.618	**0.669**
Decision Tree	0.715	0.442	0.707	**0.544**
Logistic Regression	0.601	0.352	**0.783**	0.486

Selection Rule: Model chosen based on highest F1-Score on anomalous class due to 76/24 class imbalance.

Test Set Size: Evaluated on 652 held-out images (495 normal, 157 anomalous).

ERROR ANALYSIS

Trade-Off: False Alarms vs. Missed Defects

Logistic Regression

Catches 123/157 defects but creates 226 false alarms.

```
False Alarms: 226
Missed Defects: 34
```

Decision Tree

Balanced recall but higher overall variance.

```
False Alarms: 140
Missed Defects: 46
```

Random Forest (Best)

Lowest false alarms while retaining strong defect detection.

```
False Alarms: 36
Missed Defects: 60
```

MODEL INTERPRETABILITY

Top Visual Predictors in Random Forest



Key Scientific Finding

Edge density leads because surface scratches, dents, and cracks disrupt local edge continuity.

> **Zero Category Memorization:** Object category indicator columns ranked below all visual features, proving true visual learning.

SYSTEM DEPLOYMENT

Interactive Streamlit Prototype (app.py)

Data Dashboard

Interactive visualizations of dataset splits, category shares, and edge density distributions.

Live Inference

Real-time feature extraction from image uploads or manual numeric parameter inputs.

SQLite Audit History

Automated logging of prediction probabilities, category tags, and CSV export capabilities.

```
No Retraining Required • Loads best_model_random_forest.joblib instantly
```

Streamlit v1.38+

USER WORKFLOW

Live Anomaly Prediction Protocol

1. Image Upload & Validation

Accepts PNG/JPG (≤10 MB). Validates dimensions and memory constraints.

2. Real-Time Feature Calculation

Extracts 9 visual parameters instantly without saving raw image files.

3. Plain-Language Guidance

Outputs PASS/FAIL flag with confidence score and uncertainty warnings (35–65%).

⚠ Responsible Use Protocol

Designed as a decision-support aid for human operators. High-uncertainty predictions require physical verification by a quality assurance engineer before final batch rejection.

FUTURE ARCHITECTURE

PatchCore: Pixel-Level Defect Heatmaps

Deep Feature Embeddings

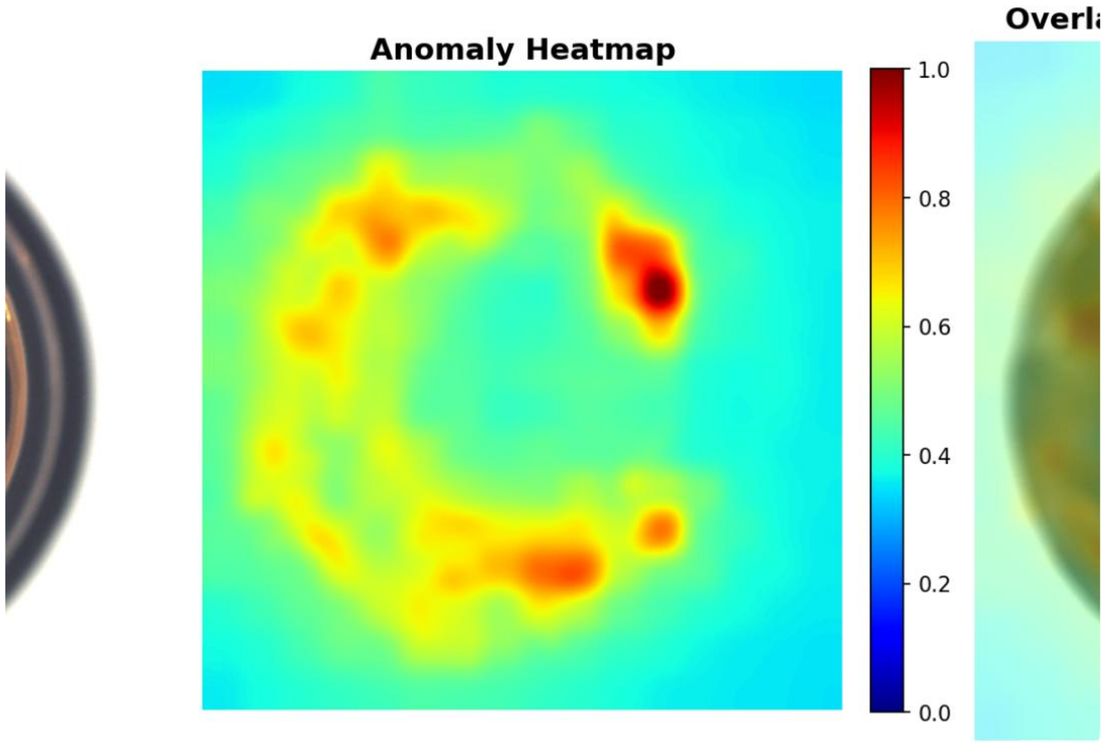
WideResNet-50 frozen backbone extracts patch neighborhood vectors.

Coreset Memory Bank

Stores only normal patch features, eliminating need for defective training data.

Distance Scoring & Heatmaps

Evaluates patch distance to generate exact pixel-level defect localization masks.



Ready for Unsupervised PatchCore Deep Feature Integration

BASELINE ACHIEVED

85.3% test accuracy & 0.669 F1-score with Random Forest classifier.

DEPLOYED PROTOTYPE

Streamlit application live with database prediction history.

Thank You — Questions & Discussion

SUZA Data Science Laboratory • Group 3 Project Presentation

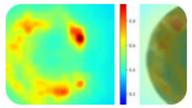
Q & A Session

Image Sources



https://ifactoryapp.com/article/manage-post-2k26/uploads/ai-vision-inspection-manufacturing-defect-detection.png

Source: ifactoryapp.com



https://cdn.prod.website-files.com/67d7ebcca3cabb623904dcde/69b9d4c15070752fa6ea051d_single_image_inference_defect.png

Source: datature.io